Ex 7.4.3

Suppose that γ is a positively oriented Jordan path in a domain D that also contains the interior of γ. Show that if $f \in H(D)$ and is nonzero on γ^*, then f has at most finitely many zeros inside of γ, and

$$N_f = \frac{1}{2\pi i}\int_\gamma \frac{f'(z)}{f(z)}\,dz$$

is the number of zeros, counted up to multiplicity, of f inside of γ. (Hint: compare with Theorem 5.35 on page 83.)

Pf. Let I be the interior of γ. By Thm 7.22 (Jordan Curve Theorem), I is a bounded simply connected domain with boundary γ^*, so we have $\bar{I} = I \cup \gamma^*$ is compact and $\bar{I} \subseteq D$.

Assume, for proof by contradiction, that f has infinitely many zeros in I. Since $\bar{I}$ is compact, the zeros have a limit point $a \in \bar{I}$. By continuity of f on D, $f(a) = 0$. The case $a \in \gamma^*$ contradicts the assumption that f is nonzero on γ^*. Therefore, $a \in I \subseteq D$ contradicts Thm 5.6, since f would have a non-isolated zero in the domain D, forcing $f \equiv 0$ on D; this contradicts f nonzero on γ^*. Therefore f has only finitely many zeros in I.

Let $a_1, \dots, a_n$ be the distinct zeros of f in I with multiplicities $m_1, \dots, m_n$. Let,

$$h(z) := \frac{f(z)}{\prod_{j=1}^n (z-a_j)^{m_j}}$$

for $z \in D \setminus \{a_1, \dots, a_n\}$. Using Lemm 5.4, near each a_k we have $f(z) = (z-a_k)^{m_k} g_k(z)$ with $g_k \in H(D)$ and $g_k(a_k) \neq 0$, therefore,

$$h(z) = \frac{g_k(z)}{\prod_{j\neq k}(z-a_j)^{m_j}}$$

near a_k, which is holomorphic with nonzero value at a_k.
Using Prop 5.13, each a_k is a removable singularity of h, and the extension (called h) lies in $H(D)$. Moreover, h is nonvanishing on $\bar{I}$: $h \neq 0$ on γ^* since $f \neq 0$ there, and $h \neq 0$ on I since the zeros of f in I have been removed in a_j.

On $D \setminus \{a_1, \dots, a_n\}$, we have,

$$\frac{f'(z)}{f(z)} = \sum_{j=1}^n \frac{m_j}{z-a_j} + \frac{h'(z)}{h(z)}$$

Let $V := \{z \in D : h(z) \neq 0\}$ be an open set containing $\bar{I}$ since h is continuous and nonzero on the compact set $\bar{I}$. Then we have, $h'/h \in H(V)$.

We can show $\mathrm{Ind}_\gamma(a) = 0\ \forall a \notin V$. Consider:

- $a \notin D$: a is in the unbounded component of $\mathbb{C} \setminus \gamma^*$ (since $\bar{I} \subseteq D$), so $\mathrm{Ind}_\gamma(a) = 0$.
- $a \in D \setminus V$: $h(a) = 0$, so $a \notin \bar{I}$ and again a is in the unbounded component, then again $\mathrm{Ind}_\gamma(a) = 0$.

Hence using Thm 6.6 (Cauchy's Theorem) on $h'/h \in H(V)$ and the cycle γ in V gives,

$$\int_\gamma \frac{h'(z)}{h(z)}\,dz = 0$$

Using Def 8.12 (since γ is positively oriented), $\forall j : \mathrm{Ind}_\gamma(a_j) = 1$. Therefore,

$$\int_\gamma \frac{f'(z)}{f(z)}\,dz = \sum_{j=1}^n m_j \int_\gamma \frac{dz}{z-a_j} + 0 = 2\pi i \sum_{j=1}^n m_j$$

Hence, $N_f = \dfrac{1}{2\pi i}\displaystyle\int_\gamma f'/f\,dz = \sum_{j=1}^n m_j$ is the number of zeros of f inside γ counted with multiplicity, as required. □

Ex 7.4.4

Prove Rouché's Theorem: suppose that f and g are analytic on and inside a positively oriented Jordan path γ and are such that $|f(z) - g(z)| < |f(z)|$ for all $z \in \gamma^*$. Show that f and g have the same number of zeros inside of γ counted up to multiplicity. (Hint: compare with Ex 6.2.3.)

Pf. Let I be the interior of γ. We should have both f and g are holomorphic on some open set D containing $\bar{I} = I \cup \gamma^*$.

For $z \in \gamma^*$, the inequality $|f(z) - g(z)| < |f(z)|$ forces $|f(z)| > 0$. Also, by the reverse of the triangle inequality, can say,

$$|g(z)| \geqslant |f(z)| - |f(z) - g(z)| > 0$$

Thus $f \neq 0$ and $g \neq 0$ on γ^*. Using Ex 7.4.3, we have the integrals,

$$N_f = \frac{1}{2\pi i} \int_\gamma \frac{f'(z)}{f(z)}\, dz$$
$$N_g = \frac{1}{2\pi i} \int_\gamma \frac{g'(z)}{g(z)}\, dz$$

which count the zeros inside γ (with multiplicity) of f and g, respectively.

Using the hint, we define $H : [0,1] \times [0,1] \to \mathbb{C}$ as,

$$H(s,t) = t\, f(\gamma(s)) + (1-t)\, g(\gamma(s))$$

which is continuous. Since $\gamma(0) = \gamma(1)$, we have: $H(s,0) = g(\gamma(s))$, $H(s,1) = f(\gamma(s))$, and $H(0,t) = H(1,t)$.
We can confirm that $H(s,t)$ is nonzero $\forall (s,t) \in [0,1]^2$. We take $H(s,t) = f(\gamma(s)) - (1-t)(f(\gamma(s)) - g(\gamma(s)))$ and usign $0 \leqslant 1 - t \leqslant 1$, we have,

$$|H(s,t)| \geqslant |f(\gamma(s))| - (1-t)|f(\gamma(s)) - g(\gamma(s))| \geqslant |f(\gamma(s))| - |f(\gamma(s)) - g(\gamma(s))| > 0.$$

Therefore, we have that H is a homotopy of closed paths in the domain $\mathbb{C} \setminus \{0\}$ between $g \circ \gamma$ (at $t = 0$) and $f \circ \gamma$ (at $t = 1$).

Using Thm 6.14, since $0 \notin \mathbb{C} \setminus \{0\}$, we have, $\mathrm{Ind}_{g\circ\gamma}(0) = \mathrm{Ind}_{f\circ\gamma}(0)$.

Using Eqn (5.15) (change of variables identity) from $(f \circ \gamma)'(s) = f'(\gamma(s))\gamma'(s)$, we have,

$$\mathrm{Ind}_{f\circ\gamma}(0) = \frac{1}{2\pi i} \int_{f\circ\gamma} \frac{dw}{w} = \frac{1}{2\pi i} \int_\gamma \frac{f'(z)}{f(z)}\, dz = N_f$$

and symmetrically $\mathrm{Ind}_{g\circ\gamma}(0) = N_g$.

Hence shown that $N_f = N_g$: f and g have the same number of zeros inside γ counted with multiplicity. □

Ex 7.4.5

> Assume the result from Chapter 8 that every Jordan path or its opposite is positively oriented (Theorem 8.1). Suppose that f is analytic on and inside a Jordan path γ. Suppose that f is one-to-one on γ so that $f(\gamma)$ is also a Jordan path. Then f maps the interior of γ bijectively onto the interior of $f(\gamma)$. In particular, f is a conformal equivalence of the two interiors and $f(\gamma)$ is positively oriented whenever γ is.

Pf. Let I be the interior of γ. We have $\bar{I} = I \cup \gamma^*$. Let D be an open set with $\bar{I} \subseteq D$ on which f is holomorphic. Since γ is piecewise smooth, closed, and simple (Jordan), and f is holomorphic and injective on γ^*, the composition $\tilde{\gamma} := f \circ \gamma$ is piecewise smooth, closed, and simple. So $\tilde{\gamma}$ is a Jordan path.
By Thm 7.22, $\mathbb{C} \setminus \tilde{\gamma}^*$ is made up of two domains: a bounded interior J and an unbounded exterior. Both have boundary, $\tilde{\gamma}^* = f(\gamma^*)$.

Take $w_0 \notin \tilde{\gamma}^*$. Then we have $f(z) - w_0$ lies in $H(D)$, and it is nonzero on γ^* (otherwise $w_0 \in f(\gamma^*)$). Using Ex 7.4.3's solution from above, we have the number of zeroes of $f - w_0$ in I (with multiplicity) as,

$$N(w_0) := \frac{1}{2\pi i} \int_\gamma \frac{f'(z)}{f(z) - w_0} \, dz$$

Then using the the same change of variables as in Ex 7.4.4's solution above, we have,

$$N(w_0) = \frac{1}{2\pi i} \int_\gamma \frac{f'(z)}{f(z) - w_0} \, dz = \frac{1}{2\pi i} \int_{\tilde{\gamma}} \frac{dw}{w - w_0} = \mathrm{Ind}_{\tilde{\gamma}}(w_0)$$

Therefore, we have $N(w_0) = \mathrm{Ind}_{\tilde{\gamma}}(w_0)$. Using Ex 6.1.1, we know that this is constant on each connected component of $\mathbb{C} \setminus \tilde{\gamma}^*$.

On the unbounded component, $\mathrm{Ind}_{\tilde{\gamma}} = 0$. Using Thm 8.1, we have on the bounded component J, the index equals ± 1, and the sign is $+1$ iff $\tilde{\gamma}$ is positively oriented. Since $N(w_0) \geqslant 0$ (it is a count of zeros), we must have $\mathrm{Ind}_{\tilde{\gamma}}(w_0) = +1$ for $w_0 \in J$. Therefore $\tilde{\gamma} = f(\gamma)$ is positively oriented.

$\forall w_0 \in J$, we have $N(w_0) = 1$. Therefore, $f - w_0$ has exactly one zero in I, counted with multiplicity. In particular, this zero is simple, hence $f'(z)$ is nonzero at that zero. Therefore, $\forall w_0 \in J$ has exactly one preimage in I, giving an injection $J \to I$ inverse to $f|_I$ restricted to $f^{-1}(J)$.

$\forall w_0$ in the unbounded component of $\mathbb{C} \setminus \tilde{\gamma}^*$, we have $N(w_0) = 0$, so $w_0 \notin f(I)$. Then we can say,

$$f(I) \subseteq J \cup \tilde{\gamma}^* = J \cup f(\gamma^*)$$

Assume, for proof by contradiction, that there is $z_0 \in I$ with $f(z_0) \in f(\gamma^*)$. Using Thm 5.38 (Open Mapping Theorem) on the nonconstant $f \in H(D)$, the image $f(I)$ is open and contains $f(z_0)$. Since $f(z_0) \in \tilde{\gamma}^* = \partial J$, any open neighborhood of $f(z_0)$ meets the unbounded component of $\mathbb{C} \setminus \tilde{\gamma}^*$. This contradicts $f(I) \subseteq J \cup \tilde{\gamma}^*$ (since J is the bounded interior). Therefore, $f(I) \subseteq J$.
Combining, f maps I onto J bijectively.

From the above, we have $f'(z)$ on I above. We also have, $f : I \to J$ is a holomorphic injection. Using Cor 5.49, f' is nonzero on I and the inverse $f^{-1} : J \to I$ is holomorphic. Then using Rmk 5.51, f is shown to be a conformal equivalence of I with J. □

Ex 10.1.2

Show that $C(K; E)$ is a subalgebra of $C(K)$.

Pf. We show $C(K; E)$ is closed under addition, scalar multiplication, and pointwise multiplication.

If $r_1, r_2 \in R(E)$, then $r_1 + r_2$ and $r_1 r_2$ are rational functions whose finite poles lie among the union of the finite poles of r_1 and r_2. Therefore, they lie in E. If either r_1 or r_2 has a pole at ∞, then we have $\infty \in E$, and the sum and product can have at most a pole at ∞, which also lies in E. Therefore, we have that: $r_1 + r_2, r_1 r_2 \in R(E)$. Similarly $\lambda r_1 \in R(E)$ for $\lambda \in \mathbb{C}$.

Let $f, g \in C(K; E)$ and $\lambda \in \mathbb{C}$. We take the sequences $(r_n), (s_n)$ in $R(E)$ with $r_n \to f$ and $s_n \to g$ uniformly on K, i.e., $\|r_n - f\|_\infty \to 0$ and $\|s_n - g\|_\infty \to 0$.

For addition and scalar multiplication, we have from the standard triangle estimates,

$$\|(r_n + s_n) - (f + g)\|_\infty \leqslant \|r_n - f\|_\infty + \|s_n - g\|_\infty \to 0$$
$$\|\lambda r_n - \lambda f\|_\infty = |\lambda| \, \|r_n - f\|_\infty \to 0$$

which gives $r_n + s_n \to f + g$ and $\lambda r_n \to \lambda f$ uniformly. Since $r_n + s_n, \lambda r_n \in R(E)$, both $f + g$ and λf lie in $C(K; E)$.

For multiplication, $C(K)$ is a Banach algebra under the supremum norm, so $\|uv\|_\infty \leqslant \|u\|_\infty \|v\|_\infty$. Since (r_n) converges, it is bounded: there is $M > 0$ with $\|r_n\|_\infty \leqslant M$ for all n. Then,

$$\|r_n s_n - fg\|_\infty \leqslant \|r_n(s_n - g)\|_\infty + \|(r_n - f)g\|_\infty \leqslant M \|s_n - g\|_\infty + \|g\|_\infty \|r_n - f\|_\infty \to 0$$

Since we have $r_n s_n \in R(E)$, the uniform limit fg must lie in $C(K; E)$.

Therefore, $C(K; E)$ is closed under addition, scalar multiplication, and pointwise multiplication. Hence it is a subalgebra of $C(K)$. □

Ex 10.1.3

Let $D = D_1(0)$ and $T = \{z \in \mathbb{C} : |z| = 1\}$ so that $\overline{D} = D \cup T$. Let $A(T)$ be the set of $f \in C(T)$ such that there is a $g \in C(\overline{D})$ with $g|_T = f$ and $g|_D \in H(D)$. (That is, $A(T)$ is the set of continuous functions on T that extend to a holomorphic function on the interior of T.) Show that $A(T)$ is a Banach subalgebra of $C(T)$. The algebra $A(T)$ is often called the disk algebra. (I suggest using Corollary 5.42 of the Maximum Modulus Theorem.)

Pf. A closed subspace of the Banach space $C(T)$ is complete in the inherited norm, hence $A(T)$ is a Banach subalgebra.

We can show that $A(T)$ is a subalgebra. Let $f_1, f_2 \in A(T)$ with extensions $g_1, g_2 \in C(\overline{D})$ that are holomorphic on D. Let $\lambda \in \mathbb{C}$. Then we have: $g_1 + g_2$, λg_1, and $g_1 g_2$ are continuous on the compact set $\overline{D}$ (sum, scalar multiple, and product of continuous functions). We also have that they are holomorphic on D using Prop 2.2 (since $H(D)$ is also closed under addition, scalar multiplication, and pointwise multiplication).
On T, we have that they restrict to $f_1 + f_2$, λf_1, and $f_1 f_2$, respectively, so all three of them lie in $A(T)$.

We can also show that $A(T)$ is closed. Let (f_n) be a sequence in $A(T)$ converging uniformly on T to $f \in C(T)$, and let $g_n \in C(\overline{D})$ be the corresponding extensions with $g_n|_D \in H(D)$. The bounded domain D, the continuous function $g_n - g_m$ on $\overline{D}$, and the holomorphic restriction $(g_n - g_m)|_D$ all fulfill the requirements of Cor 5.42 (Maximum Modulus Theorem on a bounded domain). Therefore, we have (the beautiful),

$$\max_{z \in \overline{D}} |g_n(z) - g_m(z)| = \max_{z \in T} |g_n(z) - g_m(z)| = \|f_n - f_m\|_\infty \to 0$$

as $n, m \to \infty$. Therefore, (g_n) is Cauchy in $C(\overline{D})$ under the supremum norm.
Since $C(\overline{D})$ is complete (uniform limits of continuous functions on a compact set are also continuous), $\exists g \in C(\overline{D})$ with $g_n \to g$ uniformly on $\overline{D}$.

By the estriction, $g|_T = \lim g_n|_T = \lim f_n = f$ uniformly on T. The uniform convergence $g_n \to g$ on $\overline{D}$ gives us uniform convergence on every compact subset of D. Using Thm 5.32, the limit $g|_D \in H(D)$.

Therefore, $g \in C(\overline{D})$, $g|_T = f$, and $g|_D \in H(D)$, so $f \in A(T)$. Hence $A(T)$ is closed in $C(T)$.

Therefore, we have that $A(T)$ is a closed subalgebra of the Banach algebra $C(T)$, and is hence a Banach subalgebra. □